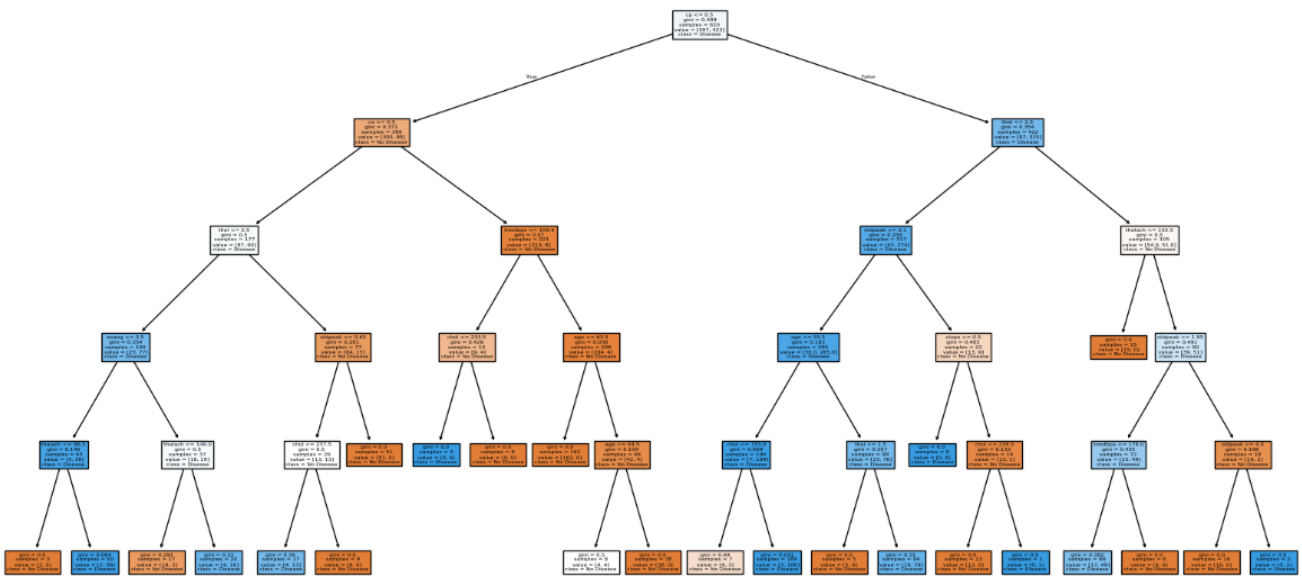
EXPERIMENT - 3

AIM: To Implement Decision Tree and Random Forest Classification algorithms on real-world datasets and evaluate their performance on the dataset - heart.csv

OUTPUT:

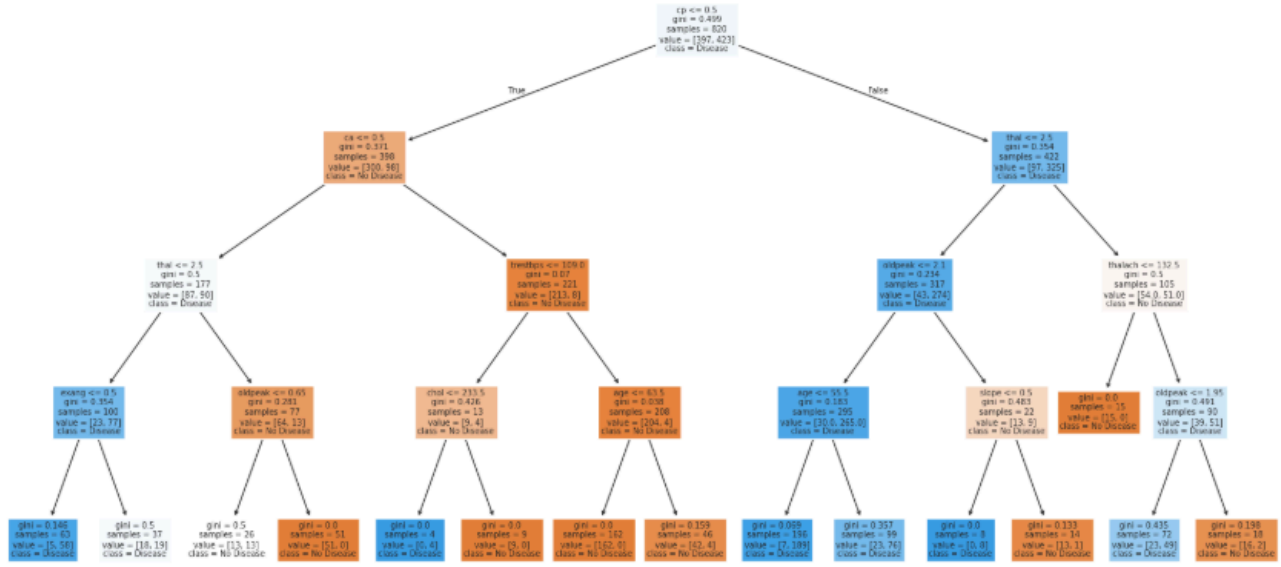
Performing decision tree algorithm (depth=5)

```
--- Decision Tree Performance ---
Accuracy: 0.84

Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.74      0.82       102
           1       0.78      0.95      0.86       103

    accuracy                           0.84       205
   macro avg       0.86      0.84      0.84       205
weighted avg       0.86      0.84      0.84       205
```

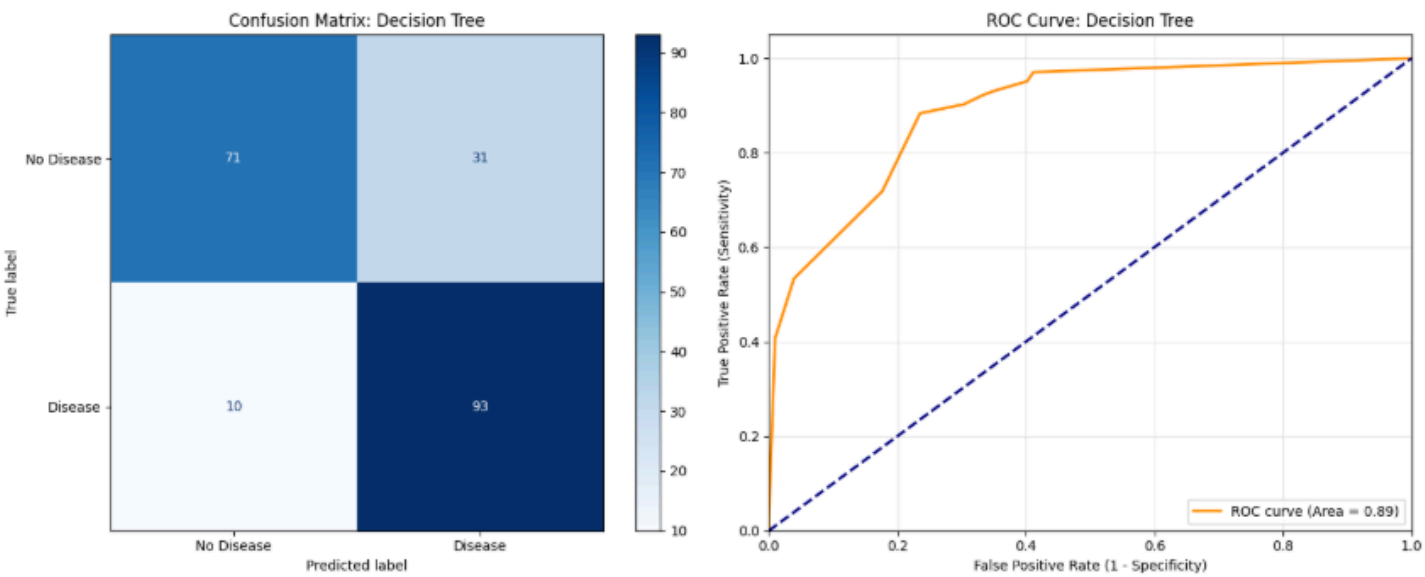


Performing decision tree algorithm (depth=4)

```
--- Decision Tree Performance ---
Accuracy: 0.80

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.70      0.78       102
           1       0.75      0.90      0.82       103

    accuracy                           0.80       205
   macro avg       0.81      0.80      0.80       205
weighted avg       0.81      0.80      0.80       205
```



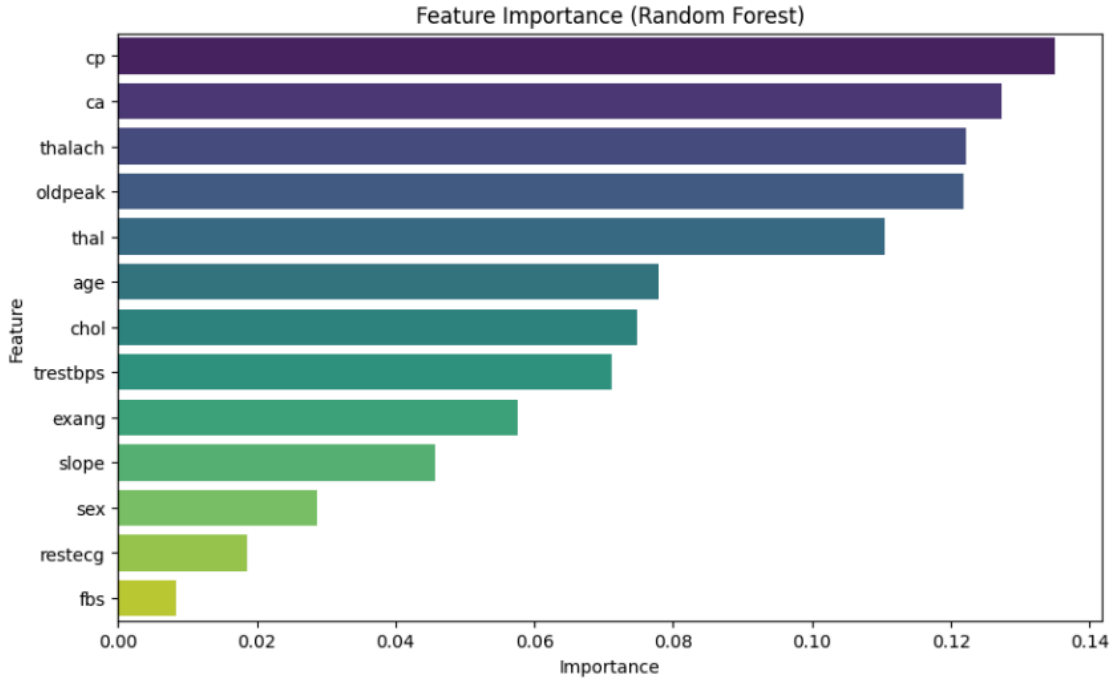
Confusion matrix and ROC-AUC Curve for *decision tree*



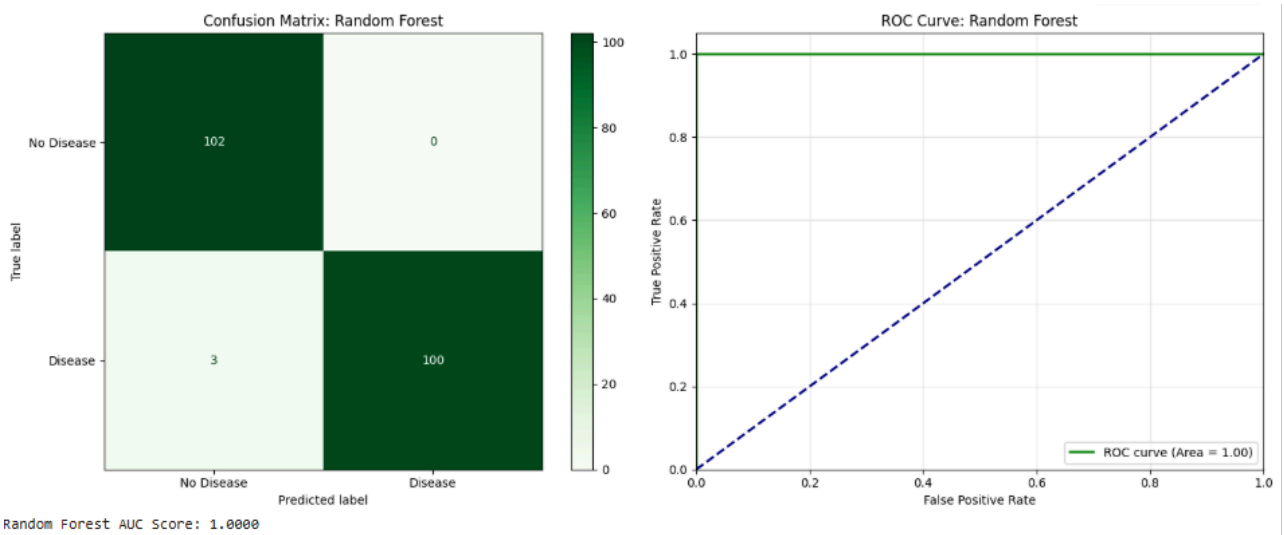
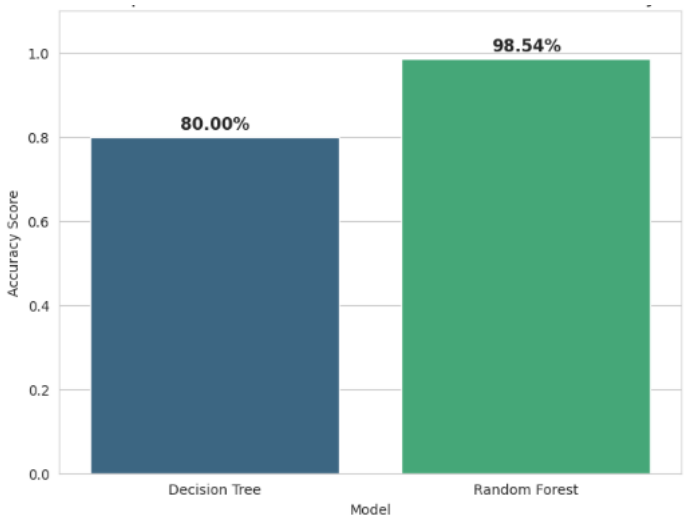

```
The Area Under the Curve (AUC) is: 0.8906
```

Random Forest

```
--- Random Forest Performance ---
Accuracy: 0.99

Classification Report:
              precision    recall  f1-score   support

           0       0.97      1.00      0.99       102
           1       1.00      0.97      0.99       103

    accuracy                           0.99       205
   macro avg       0.99      0.99      0.99       205
weighted avg       0.99      0.99      0.99       205
```



Confusion Matrix and ROC-Curve for ***Random forest***



Comparison of accuracy between Decision tree and random forest